

Coral Transplantation: Regeneration and Growth of *Acropora* Fragments in a Nursery

Keryea Soong¹ and Tai-an Chen

Abstract

Coral reef degradation has been widely reported for the past 20 years. Because the recovery rate is usually low, various methods of restoration have been explored in different regions of the world. Among the effective and commonly used methods to restore coral communities is the transplantation of coral colonies or fragments. In this investigation fragments of *Acropora pulchra* were used in a semiprotected nursery in southern Taiwan between 1996 and 1998 to test, in situ, the possible effects of different factors on the generation of new branches and the initial skeletal extension rates of transplants. The variables under study here were the origin and length of the fragments, their new orientation, presence of tissue injury, and position in the frag-

ment. All these factors were found to make a difference in either one or both aspects of coral growth (i.e., branching frequency and skeletal extension rate). These two factors clearly determine the success rate of a small fragment developing into a large colony that has a much higher probability to survive and grow on its own. It is now obvious that the efficiency of coral generation through fragment culture can be enhanced if the variables examined here are taken into consideration. Once coral colonies are formed, they can be fragmented again to generate more corals or can be transplanted to a suitable site.

Key words: *Acropora*, coral, fragment, growth, nursery, regeneration, transplantation.

Introduction

Coral reef deterioration, caused by natural and/or anthropogenic factors, has been widely reported worldwide over the last two decades (Highsmith et al. 1980; Alcala et al. 1982; Glynn 1984; Plucer-Rosario & Randall 1987; Maragos et al. 1996; Hudson & Goodwin 1997; Ormond et al. 1997; Hodgson 1999). Because of the slow natural recovery of reefs and their high economic potential, reconstruction efforts of various kinds and on a wide range of scales have been proposed and efforts have been made to facilitate reef redevelopment (Craik et al. 1990; Rinkevich 1995; Guzman 1999).

Relying on natural recruitment is one possible approach (Edwards & Clark 1998), but several limitations have been reported. First, the rate of natural recruitment of corals is often so highly variable that the process can take up to several years, especially in species broadcasting their gametes (Wallace 1985; Gleason 1996; Connell et al. 1997). Species releasing larvae (i.e., planulae) have high settlement rates in some areas, but more often than not they settle near parents and show only limited range of dispersal (Harrison & Wallace 1990). Substantial spatial variations with respect to recruitment have also been reported in many studies (Dunstan & Johnson 1998; Hughes et al. 1999); thus, many suitable habitats are too often bare due to the absence of natural recruitment. Further, coral recruits, settling on monitored surfaces, are often low in spe-

cies diversity (Harriott & Banks 1995; Smith 1997), which means a damaged reef may require a considerably long time to regain its original diversity. Another disadvantage is that recruits in nature usually suffer from high mortality and slow growth rates (Sato 1985). For example, in southern Taiwan, less than 5% of coral recruits reportedly survived 22 months, and some of those that did were still below 1 cm in diameter (Soong, unpublished data). Moreover, some destruction renders the bottom of the sea into an unconsolidated substrate that inhibits successful recruitment of corals (personal observation).

As a second approach, the transplantation of whole coral colonies has been undertaken in areas of small-scale destruction where hard substrate is still available (Maragos 1974; Hudson & Diaz 1988; Munoz-Chagin 1997). Aside from providing new colonies to the receiving sites, this procedure may also increase local recruitment as in the case when transplants are gravid (Richmond & Hunter 1990). The fixation of colonies is critical in this kind of operation, but unfortunately it too often necessitates high costs (Kaly 1995). An adequate source of live corals is also a prerequisite so that enough raw materials for transplantation can be supplied without damaging the communities of the donor sites. For obvious reasons this approach can only be realistically applied in cases of certain small-scale reconstruction.

A third approach, the transplantation of coral fragments, is also feasible (Alcala et al. 1982; Oren & Benayahu 1997; Guzman 1999). Because corals are modular organisms, small pieces of corals have the capability of growing in the same way as whole colonies (Connell 1973; Birkeland et al. 1979). The growth of corals from fragments is, in fact, an important natural process, at least in some branching species (High-

Institute of Marine Biology, National Sun Yat-sen University, Kaohsiung, 804, Taiwan.

¹Address correspondence to K. Soong, e-mail keryea@mail.nsysu.edu.tw.

smith 1982; Wallace 1985). The natural fragments first anchor and secure themselves in the crevices, more or less by chance, then continue to attach themselves to the substrate by regeneration and extension of soft tissues and skeleton. A new coral colony may start by this asexual means, and there is evidence suggesting that fragmentation is adaptive in some species (Brazeau & Lasker 1992; Fong & Lirman 1995). In artificial transplantation, exactly how to best arrange fragments, which are of course much smaller in size than whole colonies, is critical to their survival and growth rates (Guzman 1991; Custodio & Yap 1997; Oren & Benayahu 1997; Franklin et al. 1998; Rinkevich 2000). Because sedimentation often causes mortality and inhibits coral growth (Yap & Gomez 1985; Nagelkerken et al. 2000), keeping fragments above the bottom can reduce the likelihood of their being covered with silt. Placing a coral nursery structure in shallow waters may facilitate coral growth on account of the higher light intensities; nevertheless, the risk of the structures being easily damaged by strong wave actions is greatly increased (Plucer-Rosario & Randall 1987).

Strategically, it is imperative to consider all options available at the particular site being considered for reconstruction. Because no single method stands out as suitable for all situations and purposes, critical evaluations must be made before human intervention is taken (Edwards & Clark 1998). In southern Taiwan reef corals chiefly suffer from high sedimentation rates, typhoons, overfishing, macroalga proliferation, bleaching, and direct collection. Although damage is often patchy, sharp decreases in live coral coverage, species richness, species diversity, and colony numbers were recorded between 1987 and 1999 (Dai 1997; Dai et al. 1998, 1999).

Another study, undertaken for 4 years between 1997 and 2000, has indicated very low rates of natural coral recruitment in southern Taiwan, and by and large such recruits have been limited to brooding species (Soong, K., M.-H. Chen, C.-L. Chen, D.-F. Dai, T.-Y. Fan, J.-J. Li, H.-M. Fan, K.-M. Kuo, H. Hsieh. Spatial and temporal variation of coral recruitment in Taiwan, unpublished data). Thus, it may take a long time for a reef to recover if only natural recruitment is left to take its course. Using whole colonies is not practical either, because too many areas are under threat and too few healthy reefs are capable of providing transplants.

The approach we chose to study here was to use coral fragments, particularly those of the *Acropora*, because these are known to grow or extend faster than most other corals (Alcala et al. 1981; Yap & Gomez 1981). Besides making an endeavor to raise individual coral colonies, we sought to explore and assess the effectiveness of different variables in creating habitats for other corals, that is, to transform unsuitable unconsolidated substrate into hard substrate suitable for future coral recruits. Because most *Acropora* colonies had died during earlier typhoons in Taiwan, how best to efficiently exploit the surviving ones was the issue that most concerned us. In other words our objective was to determine the best procedures that would allow us to use the smallest amount of living material with-

out sacrificing much in terms of mortality or the growth rates of new fragments. Simply put, how could we propagate a large amount of acroporid corals in a relatively short time so these could later either be used to produce more corals or transplanted in whole or in part? Specifically, we wanted answers to the following questions: (1) Where should fragments from a colony be collected? (2) What is the best fragment length? (3) How should the fragments be oriented? (4) How much of a difference does tissue injury on fragments make? (5) How significant is the depth effect? (6) Is alga clearing necessary?

The skeletal extension rates and branching frequencies were used to evaluate the factors under investigation. This knowledge, we expected, would reduce the burden on source reefs while increasing the likelihood of the proliferation of new reefs.

Materials and Methods

The species studied was identified as *Acropora pulchra* (Wallace & Dai 1997; Wallace 1999). Two aspects of fragment growth were studied, namely, new branch (axial polyp) generation and extension rates of the fragments. A high rate of both would, of course, lead to fast colony formation.

Site Description

To prevent our results from being affected by physical factors, an area behind the breakwater of a power plant at Henchun in southern Taiwan (21°57.4'N, 120°50'E) was selected as our coral nursery. The site is protected from waves but has a more or less constant flow rate due to the operation of the plant. The bottom, at the 6-meter depth, is sandy, allowing the coral nursery to be easily fixed in place and maintained. There was enough light at this depth for zooxanthellate corals, because natural reefs extend down to more than 10-meter depths in the area. Guarded by the power plant, the area was less vulnerable to disturbance from tourism as well.

Racks

Racks were constructed from iron and plastic rods to keep the experimental coral fragments elevated some 30 cm above bottom and to prevent them from being smothered by sediment (Fig. 1). Then the coral fragments were attached to the horizontal rods with fishing lines to avoid predation by crawling predators, such as *Drupella* spp.

Treatment

Live fragments of *Acropora* sp. from healthy colonies were obtained using a pair of pliers. They were trimmed to the required lengths and conditions (see below) so comparisons among the branches in each treatment could be made. New axial polyp generation and length increments on both ends of the individual fragments were recorded

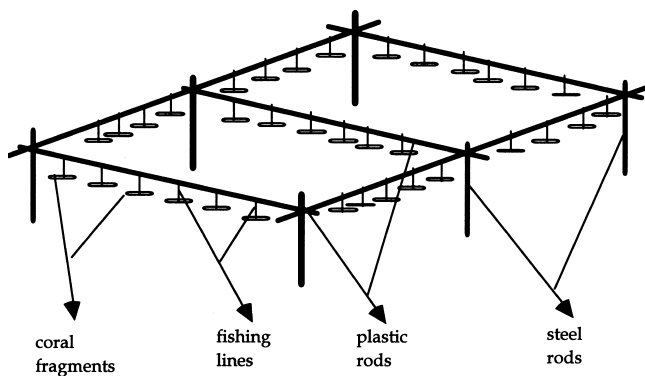


Figure 1. Coral nursery structures hung with coral fragments.

for 3 to 4 months to study the effects of fragment origin, length, new orientation, injury, and injury position on the fragment. The possible effects of alga clearing and water depth were also assessed. The original tips of the branches were cut before the experiment so that the fragments being compared all had two exposed ends in all treatments. All fragments were hung vertically unless otherwise specified.

Fragment Origin

Twenty branches were collected, and one 4-cm length fragment above and one below the most distal branching points were used to do a pair-wise comparison. Because the branches of *A. pulchra* taper toward the growing tips, the average diameters of the fragments were thicker in the proximal group, that is, those close to the base, than in the distal group.

Fragment Length

Sixty branches above the most distal branching points were used to study the effect of length among the 1-, 4- and 7-cm fragments. In this experiment shorter fragments were collected from the proximal (basal) ends of the branches; thus, longer fragments had relatively thinner diameters at their distal ends.

A further experiment involved 30, 6-cm fragments collected from the same positions in the colonies and divided into two groups: 15, 6-cm intact fragments and 15 made up of two 3-cm fragments. Because each replicate in these two groups was the same length (6 cm), they were from the same positions in the donor colonies. This was not possible in the previous experiment when different fragment lengths were compared.

Fragment Orientation

Forty 4-cm fragments were used to compare whether fragment orientation had any effect. They were hung with their original distal ends pointing either upward or downward. A second experiment of 20, 6-cm fragments hanging

horizontally was conducted to test for any possible differences with regard to the two ends being of different proximity in their donor colonies.

Injury

Artificial injuries were inflicted using a water jet to clear a 1-cm band of tissue in the center portion of the 15, 6-cm fragments. They were compared to 15 intact fragments of the same length.

Depth

All the previous experiments were conducted at the 6-meter depth. The effect of water depth itself was studied at 5- and 10-meter depths using 40, 4-cm fragments.

Alga Clearing

Filamentous algae grew on the fishing lines used to hang the coral fragments to the rods of the experimental racks. Algae were cleared away every month in the "alga clearing" treatment, whereas there was no cleaning in the control group. Both the 1- and 6-cm fragments, with 17 to 20 replicates, were used in this treatment.

Results

Fragment Origin

Almost all new branches were generated near the cut edges of the fragments (Fig. 2). The 4-cm fragments col-

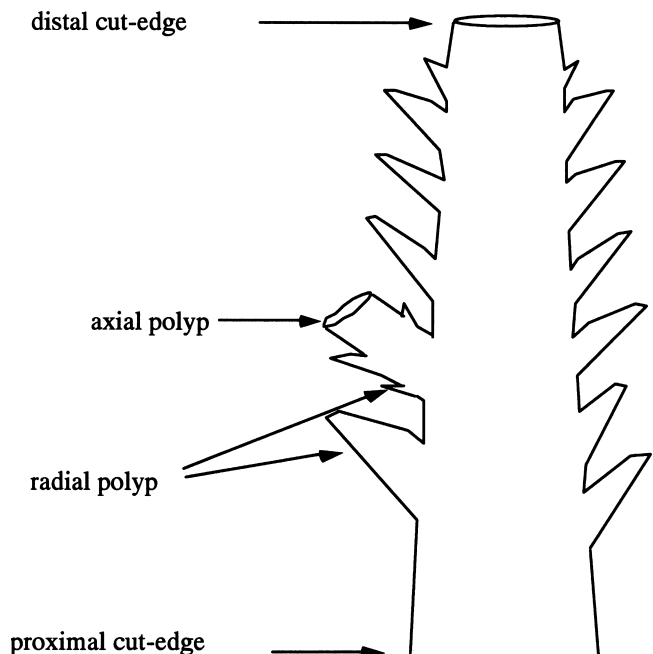


Figure 2. Morphology of fragments and the positions of new branches.

Table 1. New branch development in the 4-cm coral fragments taken from different positions along coral branches.

Fragment Origin	New Branches		p*
	Yes	No	
After 3 months			
Above the most distal bifurcation	6	13	<0.01
Below the most distal bifurcation	16	3	<0.01
After 4 months			
Above the most distal bifurcation	17	1	n.s.
Below the most distal bifurcation	18	0	n.s.

*Fisher's exact test. n.s., not significant.

lected below the most distal branching points of the colonies enjoyed a higher frequency of new branch generation (16/19 samples) than those collected above (6/19 samples, $p < 0.01$, Fisher's exact test) after 3 months. No differences whatsoever were found after 4 months, when most fragments in both groups had new branches (Table 1). The skeletal extension rates were higher in the fragments collected below rather than above the most distal branching points of the colonies after 2 and 3 months (Fig. 3).

Fragment Length

The frequencies of the fragments generating new branches were dependent on the length of the fragments. Almost all the long fragments (17/18 in the 4-cm and 18/20 in the 7-cm fragments) developed new branches within 4 months; by

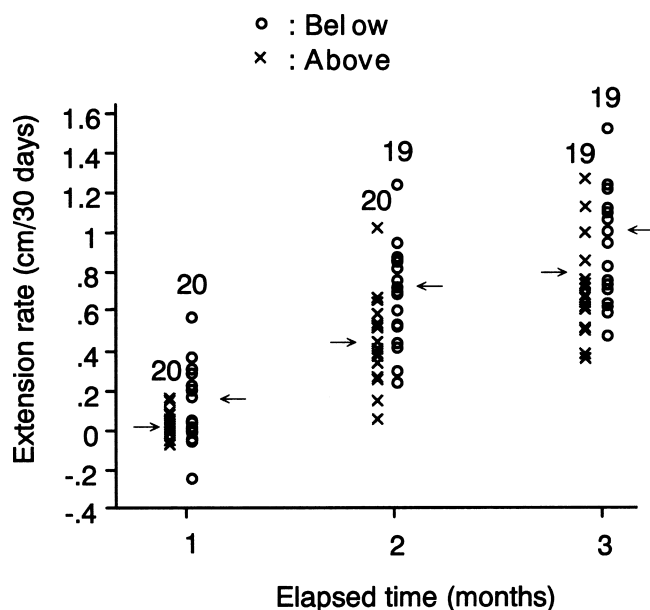


Figure 3. Skeletal extension rates of the 4-cm fragments collected below and above the most distal branching point in *Acropora* sp. colonies. Numbers indicate sample sizes. Differences were significant after 2 and 3 months ($p < 0.01$ in both cases, Wilcoxon's signed ranks tests); arrows indicate means.

Table 2. New branch development in coral fragments of different lengths.

Fragment Length (cm)	New Branches		p
	Yes	No	
	After 3 months		
1	3	14	n.s.
4	6	13	n.s.
7	8	12	n.s.
	After 4 months		
1*	8	9	
4*	17	1	
7*	18	2	

n.s., not significant.

* $p < 0.01$ between 1 cm and two other groups (Fisher's exact test).

contrast, only 8 of the 17, 1-cm fragments did so ($p < 0.01$, Fisher's exact test, Table 2).

The longer fragments had a higher skeletal extension rate than the shorter ones after 3 months. For example,

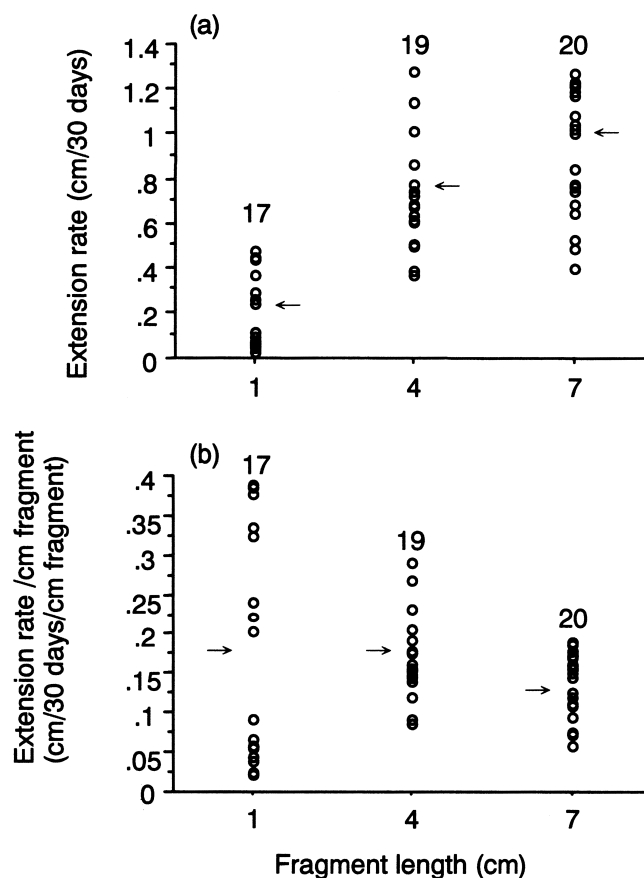


Figure 4. Skeletal extension rates of the 1-, 4-, and 7-cm fragments after 3 months. (a) Skeletal extension rate: $p < 0.01$ between the 1-cm and both the 4-cm and the 7-cm fragments; $p < 0.05$ between the 4-cm and the 7-cm fragments. (b) Skeletal extension rates per cm fragment: No significant difference was found among the three treatments. Numbers indicate sample sizes; arrows indicate means.

the 7-cm fragments increased about 1 cm in 30 days on average, whereas the 1-cm fragments increased a mere 0.2 cm during the same period (Fig. 4a). Although no significant difference was found in the specific skeletal extension rates (i.e., the extension rates per cm fragment) among the three treatments, greater variations were noticed among shorter than longer fragments. For example, the specific skeletal extension rate of the 1-cm group was between 0.02 and 0.4 cm in 30 days per fragment, a range of 20-fold difference, but that of the 7-cm group was between 0.05 and 0.2 cm in 30 days per fragment (Fig. 4b).

In the second experiment two exposed ends occurred in the 6-cm fragments, whereas a total of four exposed ends occurred in the double-3-cm fragments. There was no significant difference between the two groups in the first month, but in the second month the fastest in the 6-cm group extended 1.6 cm, yet 50% of the fragments in the double 3-cm group extended significantly more. In the third month most (11/14) in the double 3-cm group grew faster than the fastest in the 6-cm group (Fig. 5).

Fragment Orientation

Regardless of their orientation in the source colony, fragments had higher frequencies of new branch generation in the ends pointing upward than in those ends facing downward after 4 months (Table 3). It is obvious that as far as generating new branches is concerned, original proximity, whether proximal or distal, does not have as much influence as new orientation. Similarly, no difference was found between the two ends when the fragments were hung horizontally.

No difference in the total skeletal extension rates was

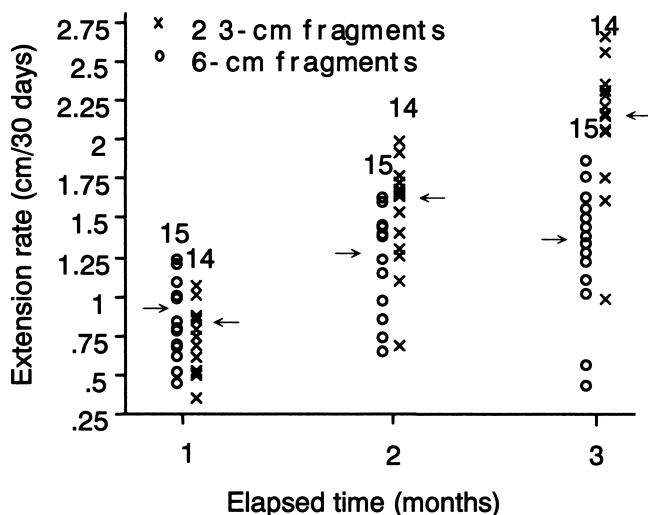


Figure 5. Overall skeletal extension rates of the 6-cm and the double 3-cm fragments from the same position along the branches of *Acropora* sp. Numbers indicate sample sizes. Differences were significant after 2 and 3 months ($p < 0.01$, Mann-Whitney U test); arrows indicate means.

Table 3. Frequencies of new branch development in coral fragments with different orientation after 4 months.

Fragment	New Branches	
	Yes	No
<i>Distal-end up</i>		
Distal end	8	7
Proximal end	1	14
<i>Distal-end down</i>		
Distal end	2	12
Proximal end	8	6

$p < 0.05$ (Fisher's exact test).

found between fragments with opposite vertical orientations ($p > 0.05$, Mann-Whitney U test). Within horizontally oriented fragments the proximal ends grew more than the distal ends. For example, only 50% of fragments from the distal ends grew less than 0.5 cm in 30 days, whereas all the proximal ends had extension rates greater than 0.5 cm in 30 days after 4 months (Fig. 6).

Injury

All exposed skeletons of fragments were covered with live coral tissue within 2 months, although the injured parts of some fragments were infested originally with algae. A lower frequency of injured fragments generated new branches than did intact fragments at the cut edges after 2 months, but this difference was no longer observed after 3 months (Table 4). However, new branches did generate between the two ends, mostly in the injured central

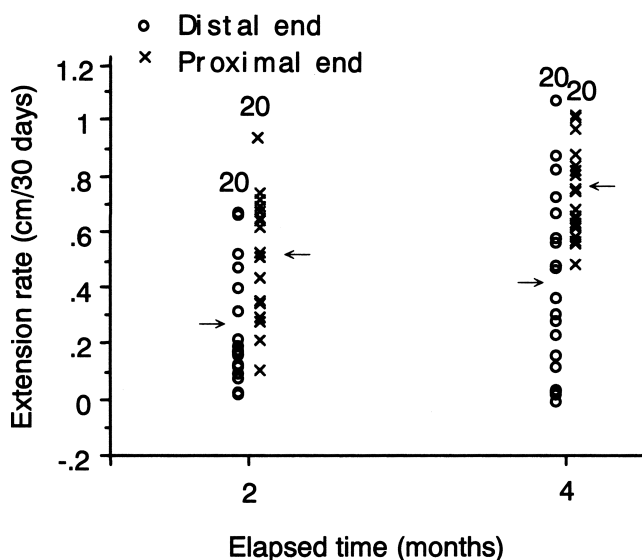


Figure 6. Skeletal extension rates of distal and proximal ends of the 6-cm horizontal fragments. Monthly differences were significant ($p < 0.01$, pair-wise Wilcoxon's signed ranks test). Numbers indicate sample sizes; arrows indicate means.

Table 4. Effect of tissue injury on the generation of new branches at the two ends of coral fragments.

Treatment	New Branches		p*
	Yes	No	
	After 2 months		
Injured	0	13	<0.01
Noninjured	9	6	<0.01
	After 3 months		
Injured	4	9	n.s.
Noninjured	9	6	n.s.

*Fisher's exact test. n.s., not significant.

parts, in 7 of the 13 fragments. This compares with only 1 of the 14 intact fragments that developed new branches in the central part of the fragments 3 months after injury (Table 5).

Lower extension rates of injured fragments relative to those of intact fragments were observed from the first to the third months. For instance, 10 of the 15 intact fragments had higher skeletal extension rates than did even the fastest injured fragments (Fig. 7).

Depth

A higher frequency of fragments at the 5-meter depth generated new branches than those at 10 meters after 4 months (18/20 vs. 8/20; $p < 0.01$, Fisher's exact test; Table 6). Similarly, significantly higher skeletal extension rates occurred at 5 meters than at 10 meters after 2 months (Fig. 8).

Alga Clearing

In the 1-cm fragments the survival rates depended on alga clearing (17/17 with clearing vs. 10/17 without clearing; $p < 0.01$, Fisher's exact test). No such result was found among the 6-cm fragments (18/18 vs. 19/20; Table 7), where the survival rates were high in both groups after 4 months. No significant differences in the skeletal extension rates were found between survivors with or without alga clearing ($p > 0.05$, Mann-Whitney U test).

Table 5. Effect of tissue injury on the development of new branches in the central part of coral fragments.

	<i>New Branches</i>		
<i>Treatment</i>	<i>Yes</i>	<i>No</i>	p*
	<i>After 2 months</i>		
Injured	5	8	n.s.
Noninjured	1	14	n.s.
	<i>After 3 months</i>		
Injured	7	6	<0.05
Noninjured	1	14	<0.05

*Fisher's exact test. n.s., not significant.

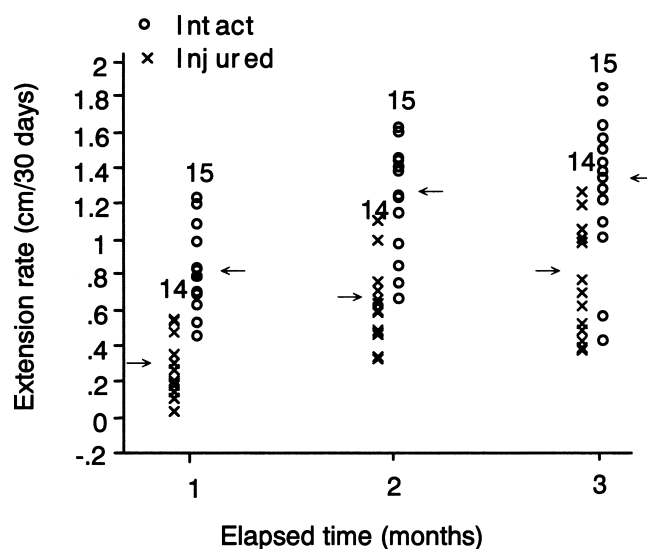


Figure 7. Skeletal extension rates of intact and injured 6-cm fragments. Numbers indicate sample sizes. Monthly differences were significant ($p < 0.01$, Mann-Whitney U tests); arrows indicate means.

Discussion

A coral fragment can escape the high risk of mortality after reaching a certain size (Connell 1973; Highsmith 1982). Critical to survival are the skeletal extension rates and branching frequencies in the initial period after the transplantation of fragments. Faster growth is particularly important, especially for small fragments. Equally important is branching frequencies because a fragment with branches can stabilize on the bottom, thereby making it possible to attach and secure itself to the substrate easily. Apart from this, they protrude beyond the bottom layer and are less likely to be totally covered with sediment or other encrusting organisms (Jackson 1985). A fragment with many branches can also extend in various directions, which raises its potential for high total growth.

The origin of the fragments seems to have made a difference in terms of both the skeletal extension rates and branching frequencies in our experiments. Fragments collected from basal or the more proximal part of donor colo-

Table 6. New branch development in coral fragments transplanted to different depths.

Depth (m)	New Branches		p*
	Yes	No	
	After 3 months		
5	9	11	n.s.
10	3	14	n.s.
	After 4 months		
5	18	2	<0.01
10	8	9	<0.01

*Fisher's exact test. n.s., not significant.

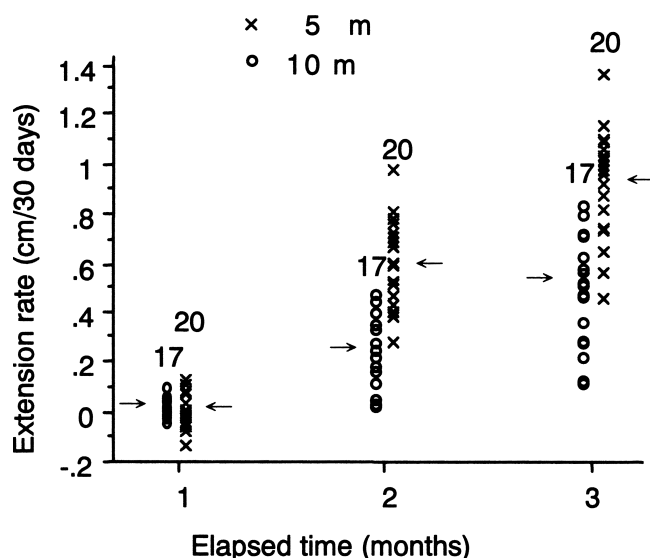


Figure 8. Skeletal extension rates of the 4-cm fragments at different depths. Differences were significant after 2 months ($p < 0.01$, Mann-Whitney U test). Numbers indicate sample sizes; arrows indicate means.

nies had thicker diameters and, as a consequence, more surface area and live tissues per unit length of fragments. Their ability to acquire energy, either by capturing food particles or by photosynthesis, is potentially higher than thin fragments that have much less surface area. Skeleton porosity also increases toward the distal ends of acroporid branches (Gladfelter 1982), perhaps allowing more energy being spent in local calcification in distal branches. Possible too is that thick branches in the basal area of donor colonies have more nutrients to translocate for the linear extension of branches (Gladfelter 1983; Fang et al. 1989). Regardless of the possible mechanisms at play, the phenomenon above creates a dilemma in transplanting fragments if the number of donor colonies is limited. Fragments collected from the proximal part of *Acropora* colonies are better from the perspective of fragment growth and branching. However, removing basal parts, at the same time, means destabilizing more distal branches in

the donor colonies. Nevertheless, under certain conditions it is possible to remove proximal branches without destroying distal branches, especially from the periphery of large *Acropora* colonies. In some habitats proximal branches in the basal part of donor colonies are likely to be encrusted and suffocated by sea anemones in our study area. These branches do, however, extend and branch after being transplanted to a well lit and less crowded environment. In light of this it is possible to attain higher total growth if proximal branches are removed from the original environment and transplanted to a new location.

Comparisons of fragment length indicate that very small fragments (e.g., 1 cm) were too small to use for coral generation in nurseries, because they tend to be smothered by algae or simply get lost, perhaps on account of predation. Moreover, in this study their branching frequencies were low, implying that their potential for propagation purposes is low. Small fragments, nevertheless, may be considered if labor is not a limiting factor and if algae can be cleared frequently (e.g., in a land-based laboratory). If this is the case, we suggest that branches of intermediate diameters be used. Thick branches of *Acropora* tend to have cores without live tissues; fouling organisms, such as algae, once growing and establishing themselves on the exposed ends of these surfaces, are likely to inhibit further growth of the corals. Additionally, from the perspective of the donor colonies, removal of long branches may render the original colonies infertile or result in low fecundities for some years (Lirman 2000).

Acropora fragments about 4 cm long are considered the most appropriate for use in coral generation, because their branching frequency in this study was no less than that of the longer fragments. In fact, 4-cm fragments might be preferred when the overall skeletal extension rate is taken into account. Because new branches were generated on the cut edges of the fragments, it is also recommended that longer branches are broken into 4-cm fragments to produce more cut edges per unit length of fragments.

Not unlike new generation on the cut edges, new branches were generated near the edge of injuries where living tissue was washed away by strong water jet and only the skeleton remained intact. Whereas position effect of coral regenerative capability has been reported in some earlier studies (Meesters & Bak 1995; Hall 1997), the preferential generation of axial polyps at certain parts of fragments has yet to receive its due attention. The location of new axial polyps may be affected by two factors. The first is developmental constraint. In our observations axial polyps could be formed only de novo in *A. pulchra* and not by the transformation of existing radial polyps. An axial polyp may become degenerated and resembles a radial polyp in the colonies of the branching *Acropora formosa* (Oliver 1984). Such a branch (i.e., without an axial polyp) had lower skeletal extension rates than others with an axial polyp at the branch tips. Investigations into the transformation from a radial to an axial polyp have yet to be reported in the available literature.

Table 7. Effect of alga clearing on the mortalities of coral fragments with different lengths after 4 months.

	<i>Dead</i>		
<i>Treatment</i>	<i>Yes</i>	<i>No</i>	p*
	<i>1 cm</i>		
Alga clearing	0	17	<0.01
No clearing	7	10	<0.01
	<i>6 cm</i>		
Alga clearing	0	18	n.s.
No clearing	1	19	n.s.

*Fisher's exact test. n.s., not significant.

Resources allotted to regeneration are limited (Meesters & Bak 1995; Hall 1997; Rinkevich 1997); this may well be the second factor influencing where to develop new axial polyps. The finding that only certain new tissues generate new axial polyps in a fragment supports the notion of a resource limitation. In short, not all parts of a fragment have the ability to develop new axial polyps. Moreover, the results also suggest that the direction of nutrient translocation in an acroporid branch can be flexible and reversed (Gladfelter 1983). Axial polyps may be generated near the cut edges, if not for the presence of tissue wound in the center portions of fragments. Thus, the original direction of translocation (i.e., toward the branch tip) had to be reversed in half of the branches when the new axial polyps were generated in the center portion of the fragments. A similar deduction can be reached for intact fragments that generate new axial polyps at both ends of the fragments. Given the position effect of axial polyp generation, artificial wounds may be strategically located on long branches to induce axial polyps where new branches are wanted.

Branching acroporids are known to translocate nutrients directionally, which leads to faster extension rates of axial polyps (Gladfelter 1983; Fang et al. 1989). Likewise, the ability of corals to regenerate was found to be dependent on the position of the injuries in the colonies. In *Acropora palmata* the regenerative capability decreases away from the growing edge (Meesters & Bak 1995). In a multispecies comparison, however, no position effect was found in the regenerative ability in six of seven species (Hall 1997). The results of our orientation experiments on fragments without axial polyps, however, indicate that the distal or proximal ends in the original colony did not have any inherent advantage in generating new axial polyps. Instead, the local environment determined the end at which new axial polyps were produced. It is possible that all the branches we used in the experiment were distal branches of the colonies and that the two ends of the 6-cm fragments posed little difference in ontogenic gradients along the branches. Accordingly, whichever end pointed upward had a higher frequency of generating new axial polyps. It may be concluded that the resulting new branches are likely to be distributed in the upper portions of fragments. This characteristic is potentially adaptive in that branches in lower shaded regions of a colony tend to be overgrown and smothered by other organisms or sediments (Meesters et al. 1997).

Light intensity is obviously an important factor affecting the extension rates of *Acropora*. Coral fragments, like whole colonies, extend faster in shallow water than in deeper waters (Huston 1985; Custodio & Yap 1997; Nagelkerken et al. 1999). On the other hand, fragments on racks in shallow waters (5 meters) were more easily destroyed by waves than those in deeper waters (10 meters). Similar results have been observed in nonbranching species where deep (10-meter) transplants survived better than shallow ones at 1.5 meters (Plucer-Rosario & Randall 1987). Simply put, the structure of the racks must be rein-

forced to sustain wave actions in shallow waters of an exposed environment, or, alternatively, a compromise has to be made between potential growth rates and the risk of survival.

In summary, our study indicated that the success of our coral fragments in a semiprotected nursery depended on many factors. With all factors taken into consideration, a large amount of acroporid corals could be produced within a reasonable period. These materials can then be used either to restore natural populations directly or to satisfy the market demand for live corals, which would obviously reduce exploitation of natural populations.

Acknowledgments

We thank Ming-hui Chen, Yung-tse Chen, Hei Jiang, Junren Cao, Dan Chang, Hon-ming Fan, Jun-yao Fu, and Yawen Young for helping with the field work. This study was supported by grants from the Taipower Company and the National Science Council (NSC89-2611-B-110-001) of Taiwan, Republic of China.

LITERATURE CITED

- Alcala, A. C., L. C. Alcala, E. D. Gomez, M. E. Cowan, and H. T. Yap. 1981. Growth of certain corals, molluscs and fish in artificial reefs in the Philippines. *Proceedings of the 4th International Coral Reef Symposium, Manila* 2:215–220.
- Alcala, A. C., E. D. Gomez, and L. C. Alcala. 1982. Survival and growth of coral transplants in central Philippines. *Kalikasan Philippine Journal of Biology* 11:136–147.
- Birkeland, C., R. H. Randall, and G. Grimm. 1979. Three methods of coral transplantation for the purpose of re-establishing a coral community in the thermal effluent area of the Tanguisson Power Plant. Technical Report 60. University of Guam Marine Laboratory, Guam.
- Brazeau, D. A., and H. R. Lasker. 1992. Growth rates and growth strategy in a clonal marine invertebrate, the Caribbean octocoral *Briareum asbestinum*. *Biological Bulletin* 183:269–277.
- Connell, J. E. 1973. Population ecology of reef building corals. Pages 205–245 in O. A. Jones and R. Endeans, editors. *Biology and geology of coral reefs*. Academic Press, London.
- Connell, J. H., T. P. Hughes, and C. C. Wallace. 1997. A 30-year study of coral abundance, recruitment, and disturbance at several scales in space and time. *Ecological Monograph* 67:461–488.
- Craik, W., R. Kenchington, and G. Kelleher. 1990. Coral reef management. Pages 453–467 in Z. Dubinskys, editor. *Coral reefs*. Elsevier, New York.
- Custodio, H. M. I., and H. T. Yap. 1997. Skeletal extension rates of *Porites cylindrica* and *Porites (Synaraea) rus* after transplantation to two depths. *Coral Reefs* 16:267–268.
- Dai, C. F. 1997. Assessment of the present health of coral reefs in Taiwan. Pages 123–131 in R. W. Grigg and C. Birkelands, editors. *Status of coral reefs in the Pacific*. Sea Grant Program, University of Hawaii, Honolulu.
- Dai, C. F., Y. T. Chen, K. M. Kuo, and C. H. Chuang. 1998. Changes of coral communities in Nanwan Bay, Kenting National Park: 1987–1997. *Journal of National Park (Taiwan, ROC)* 8:79–99 (in Chinese with English abstract).
- Dai, C. F., K. M. Kuo, Y. T. Chen, and C. H. Chuang. 1999. Changes of coral communities on the east and west coast of the Kenting National Park. *Journal of National Park (Taiwan, ROC)* 9:112–130 (in Chinese with English abstract).

- Dunstan, P. K., and C. R. Johnson. 1998. Spatio-temporal variation in coral recruitment at different scales on Heron Reef, southern Great Barrier Reef. *Coral Reefs* **17**:71–81.
- Edwards, A. J., and S. Clark. 1998. Coral transplantation: a useful management tool or misguided meddling? *Marine Pollution Bulletin* **37**: 474–487.
- Fang, L. S., Y. W. J. Chen, and C. S. Chen. 1989. Why does the white tip of stony coral grow so fast without zooxanthellae? *Marine Biology* **103**: 359–363.
- Fong, P., and D. Lirman. 1995. Hurricanes cause population expansion of the branching coral *Acropora palmata* (Scleractinia)—wound-healing and growth-patterns of asexual recruits. *Marine Ecology-Pubblicazioni Della Stazione Zoologica di Napoli* **16**:317–335.
- Franklin, H., C. A. Muhando, and U. Lindahl. 1998. Coral culturing and temporal recruitment patterns in Zanzibar, Tanzania. *Ambio* **27**: 651–655.
- Gladfelter, E. H. 1982. Skeletal development in *Acropora cervicornis*. 1. Patterns of calcium carbonate accretion in the axial corallite. *Coral Reefs* **1**:45–51.
- Gladfelter, E. H. 1983. Circulation of fluids in the gastrovascular system of the reef coral *Acropora cervicornis*. *Biological Bulletin* **165**:619–636.
- Gleason, M. G. 1996. Coral recruitment in Moorea, French Polynesia: the importance of patch type and temporal variation. *Journal of Experimental Marine Biology and Ecology* **207**:79–101.
- Glynn, P. W. 1984. Widespread coral mortality and the 1982–83 El Niño warming event. *Environmental Conservation* **11**:133–146.
- Guzman, H. M. 1991. Restoration of coral reefs in Pacific Costa Rica. *Conservation Biology* **5**:189–195.
- Guzman, H. M. 1999. Large-scale restoration of Eastern Pacific reefs: the need for understanding regional biological processes. National Coral Reef Institute, Nova Southeastern University, Fort Lauderdale, Florida.
- Hall, V. R. 1997. Interspecific differences in the regeneration of artificial injuries on scleractinian corals. *Journal of Experimental Marine Biology and Ecology* **212**:9–23.
- Harriott, V. J., and S. A. Banks. 1995. Recruitment of scleractinian corals in the Solitary Islands Marine Reserve, a high latitude coral-dominated community in eastern Australia. *Marine Ecology Progress Series* **123**:155–161.
- Harrison, P. L., and C. C. Wallace. 1990. Reproduction, dispersal and recruitment of scleractinian corals. Pages 133–207 in Z. Dubinskys, editor. *Ecosystems of the world*. 25. Coral reefs. Elsevier, New York.
- Highsmith, R. C. 1982. Reproduction by fragmentation in corals. *Marine Ecology Progress Series* **7**:207–226.
- Highsmith, R. C., A. C. Riggs, and C. M. D'Antonio. 1980. Survival of hurricane-generated coral fragments and a disturbance model of reef calcification/growth rates. *Oecologia* **46**:322–329.
- Hodgson, G. 1999. A global assessment of human effects on coral reefs. *Marine Pollution Bulletin* **38**:345–355.
- Hudson, J. H., and R. Diaz. 1988. Damage survey and restoration of M/V Wellwood Grounding Site, Molasses Reef, Key Largo National Marine Sanctuary, Florida. Proceedings of 6th International Coral Reef Symposium, Australia **2**:231–236.
- Hudson, J. H., and W. B. Goodwin. 1997. Restoration and growth rate of hurricane damaged pillar coral (*Dendrogyra cylindrus*) in the Key Largo National Marine Sanctuary, Florida. Proceedings of 8th International Coral Reef Symposium, Panama **1**:567–570.
- Hughes, T. P., A. H. Baird, E. A. Dinsdale, N. Moltschaniswkyj, M. S. Pratchett, J. E. Tanner, and B. Willis. 1999. Patterns of recruitment and abundance of corals along the Great Barrier Reef. *Nature* **397**: 59–63.
- Huston, M. 1985. Variation in coral growth rates with depth at Discovery Bay, Jamaica. *Coral Reefs* **4**:19–25.
- Jackson, J. B. C. 1985. Distribution and ecology of clonal and asexual benthic invertebrates. Pages 297–355 in J. B. C. Jackson, L. W. Buss, and R. E. Cooks, editors. *Population biology and evolution of clonal organisms*. Yale University Press, New Haven, Connecticut.
- Kaly, U. L. 1995. Experimental test of the effects of methods of attachment and handling on the rapid transplantation of corals. CRC Reef Research Center Technical Report **1**:1–24.
- Lirman, D. 2000. Fragmentation in the branching coral *Acropora palmata* (Lamarck): growth, survivorship, and reproduction of colonies and fragments. *Journal of Experimental Marine Biology and Ecology* **251**:41–57.
- Maragos, J. E. 1974. Coral transplantation: a method to create, preserve, and manage coral reefs. Sea Grant Advisory Report, UNHI-SEA GRANT-AR-74-03 CORMAR-14. University of Hawaii, Honolulu.
- Maragos, J. E., M. P. Crosby, and J. W. McManus. 1996. Coral reefs and biodiversity: a critical and threatened relationship. *Oceanography* **9**: 83–99.
- Meesters, E. H., and R. P. M. Bak. 1995. Age-related deterioration of a physiological function in the branching coral *Acropora palmata*. *Marine Ecology Progress Series* **121**:203–209.
- Meesters, E. H., I. Wesseling, and R. P. M. Bak. 1997. Coral colony tissue-damage in 6 species of reef-building corals—partial mortality in relation with depth and surface-area. *Journal of Sea Research* **37**:131–144.
- Munoz-Chagin, R. F. 1997. Coral transplantation program in the Paraiso Coral Reef, Cozumel Island, Mexico. Proceedings of 8th International Coral Reef Symposium **2**:2075–2078.
- Nagelkerken, I., S. Bouma, S. Vandenakker, and R. P. M. Bak. 2000. Growth and survival of unattached *Madracis mirabilis* fragments transplanted to different reef sites, and the implication for reef rehabilitation. *Bulletin of Marine Science* **66**:497–505.
- Nagelkerken, I., E. H. Meesters, and R. P. M. Bak. 1999. Depth-related variation in regeneration of artificial lesions in the Caribbean corals *Porites astreoides* and *Stephanocoenia michelinii*. *Journal of Experimental Marine Biology and Ecology* **234**:29–39.
- Oliver, J. K. 1984. Intra-colony variation in the growth of *Acropora formosa*: extension rates and skeletal structure of white (zooxanthellae free) and brown-tipped branches. *Coral Reefs* **3**:139–147.
- Oren, U., and Y. Benayahu. 1997. Transplantation of juvenile corals: a new approach for enhancing colonization of artificial reefs. *Marine Biology* **127**:499–505.
- Ormond, R., O. Hassan, D. Medio, M. Pearson, and M. Salem. 1997. Effectiveness of coral protection programmes in the Ras Mohammed National Park, Egyptian Red Sea. Proceedings of 8th International Coral Reef Symposium **2**:1931–1936.
- Plucer-Rosario, G., and R. H. Randall. 1987. Preservation of a rare coral species by transplantation and examination of their recruitment and growth. *Bulletin of Marine Science* **41**:585–593.
- Richmond, R. H., and C. L. Hunter. 1990. Reproduction and recruitment of corals: comparisons among the Caribbean, the Tropical Pacific and the Red Sea. *Marine Ecology Progress Series* **60**:185–203.
- Rinkevich, B. 1995. Restoration strategies for coral reefs damaged by recreational activities: the use of sexual and asexual recruits. *Restoration Ecology* **3**:241–251.
- Rinkevich, B. 1997. Do reproduction and regeneration in damaged corals compete for energy allocation? *Marine Ecology Progress Series* **143**: 297–302.
- Rinkevich, B. 2000. Steps towards the evaluation of coral reef restoration by using small branch fragments. *Marine Biology* **136**:807–812.
- Sato, M. 1985. Mortality and growth of juvenile coral *Pocillopora damicornis* (Linnaeus). *Coral Reefs* **4**:27–33.
- Smith, S. R. 1997. Patterns of coral settlement, recruitment and juvenile mortality with depth at Conch Reef, Florida. Proceedings of 8th International Coral Reef Symposium **2**:1197–1202.

- Wallace, C. C. 1985. Reproduction, recruitment and fragmentation in nine sympatric species of the coral genus *Acropora*. *Marine Biology* **88**: 217–233.
- Wallace, C. C. 1999. Staghorn corals of the world. A revision of the coral genus *Acropora* (Scleractinia; Astrocoeniina; Acroporidae) world-wide, with emphasis on morphology, phylogeny and biogeography. CSIRO Publishing, Collingwood, Victoria, Australia.
- Wallace, C. C., and C.-F. Dai. 1997. Scleractinia of Taiwan IV. Review of the coral genus *Acropora* from Taiwan. *Zoological Studies* **36**:288–324.
- Yap, H. T., and E. D. Gomez. 1981. Growth of *Acropora pulchra* (Brook) in Bolinao, Pangasinan, Philippines. *Proceedings of the 4th International Coral Reef Symposium, Manila* **2**:205–213.
- Yap, H. T., and E. D. Gomez. 1985. Growth of *Acropora pulchra*. III. Preliminary observations on the effects of transplantation and sediment on the growth and survival of transplants. *Marine Biology* **87**:203–209.